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Introduction to Computer Science

Exam Code: CS101

Date: Semester 2, 2026

Duration: 2 hours 30 minutes

Total Marks: 70

Department of Computer Science, University of OmniLaTeX

Instructions:

Answer **all** questions in the spaces provided. This is a closed-book examination. No electronic devices, notes, or textbooks are permitted. Calculators are allowed for Questions 3 and 4 only. *Answer*

all questions. Show all working where applicable.

Question 1 [5 marks]

Define the term **algorithm** and explain the difference between *time complexity* and *space complexity*. Give a brief example of each.

Question 2 [5 marks]

Which of the following data structures is most appropriate for implementing a breadth-first search (BFS)?

1. Stack
2. Queue
3. Heap
4. Binary search tree

Justify your answer in one sentence.

Question 3 [10 marks]

Let $f(n) = 3n^2 + 5n + 7$. Show that $f(n) = O(n^2)$ using the formal definition of Big-O notation. *Hint:* Find constants $c > 0$ and $n_0 \geq 1$ such that $f(n) \leq c \cdot n^2$ for all $n \geq n_0$.

Question 4 [20 marks]

Prove by mathematical induction that for all integers $n \geq 1$:

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}.$$

- (a) State the base case and verify it. [5 marks]
(b) State the inductive hypothesis. [5 marks]
(c) Complete the inductive step. [10 marks]

Question 5 [15 marks]

Consider the following recurrence relation describing the running time of a divide-and-conquer algorithm:

$$T(n) = 2T\left(\frac{n}{2}\right) + n, \quad T(1) = 1.$$

Using the Master Theorem, determine the asymptotic running time $T(n)$. Show all steps of your reasoning, including identification of the parameters a , b , and $f(n)$, and the comparison of $f(n)$ with $n^{\log_b a}$.

Question 6 [15 marks]

(a) Describe the difference between a **min-heap** and a **max-heap**. [5 marks]

(b) Given the sequence $\langle 25, 17, 36, 2, 8, 19, 45 \rangle$, draw the min-heap that results after inserting these elements one by one (in the order given). [10 marks]